

HR WORKFORCE ATTRITION ANALYSIS

1. Executive Summary

Employee attrition is one of the most significant workforce challenges faced by organizations. High employee turnover results in increased recruitment costs, loss of organizational knowledge, reduced productivity, and lower employee morale.

This project aims to analyze employee workforce data and identify patterns associated with employee attrition. Interactive Power BI dashboards were developed to monitor workforce health, employee satisfaction, attrition trends, and risk factors. The analysis provides actionable insights that can support HR leaders in making data-driven retention decisions.

2. Introduction

Human Resource Analytics is the process of collecting, analyzing, and interpreting employee-related data to improve workforce management and organizational performance.

Organizations today face increasing challenges in retaining skilled employees. Understanding why employees leave and identifying high-risk employee groups is essential for maintaining a stable and productive workforce.

This project focuses on developing a comprehensive HR Workforce Attrition Dashboard that helps management understand workforce behavior and implement effective retention strategies.

3. Problem Statement

A growing technology company has experienced a significant increase in employee turnover over the last year. Rising attrition has created operational disruptions and increased hiring costs.

Management requires an analytics solution to answer the following questions:

- Which factors contribute most to employee attrition?

- Which departments experience higher turnover?
- Does overtime affect employee retention?
- How does work-life balance influence attrition?
- Which employee groups are at higher risk?

The objective is to build an interactive dashboard capable of identifying workforce trends and supporting strategic HR decisions.

4. Business Objectives

The primary objectives of this project are:

Workforce Analysis

- Understand employee demographics.
- Analyze workforce distribution across departments.

Attrition Analysis

- Identify departments with high turnover.
- Investigate factors contributing to attrition.

Employee Health Analysis

- Measure satisfaction levels.
- Evaluate work-life balance.
- Assess overtime impact.

Retention Strategy Support

- Identify high-risk employees.
- Provide recommendations to improve retention.

5. Dataset Description

The dataset contains employee-related information used for workforce and attrition **analysis**.

Key Attributes

Category	Attributes
Employee Information	Employee ID, Gender, Marital Status
Organization Details	Department, Job Role
Workforce Metrics	Experience Level, Tenure
Satisfaction Metrics	Job Satisfaction, Environment Satisfaction
Health Indicators	Work-Life Balance, Overtime
Performance Metrics	Performance Rating
Attrition Indicators	Attrition Status

Dataset Characteristics

- Workforce records across multiple departments
- Employee engagement metrics
- Performance evaluation metrics
- Workforce health indicators

6. Data Cleaning & Preparation

Several preprocessing steps were performed to ensure data quality.

Data Cleaning Activities

Missing Value Handling

Missing records were identified and treated appropriately.

Duplicate Removal

Duplicate employee records were removed.

Data Validation

Data types and value ranges were verified.

Derived Variables Created

Experience Group

- 0–2 Years
- 3–5 Years

- 6–10 Years
- 10+ Years

Satisfaction Groups

- Job Satisfaction Group
- Environment Satisfaction Group
- Work-Life Balance Group

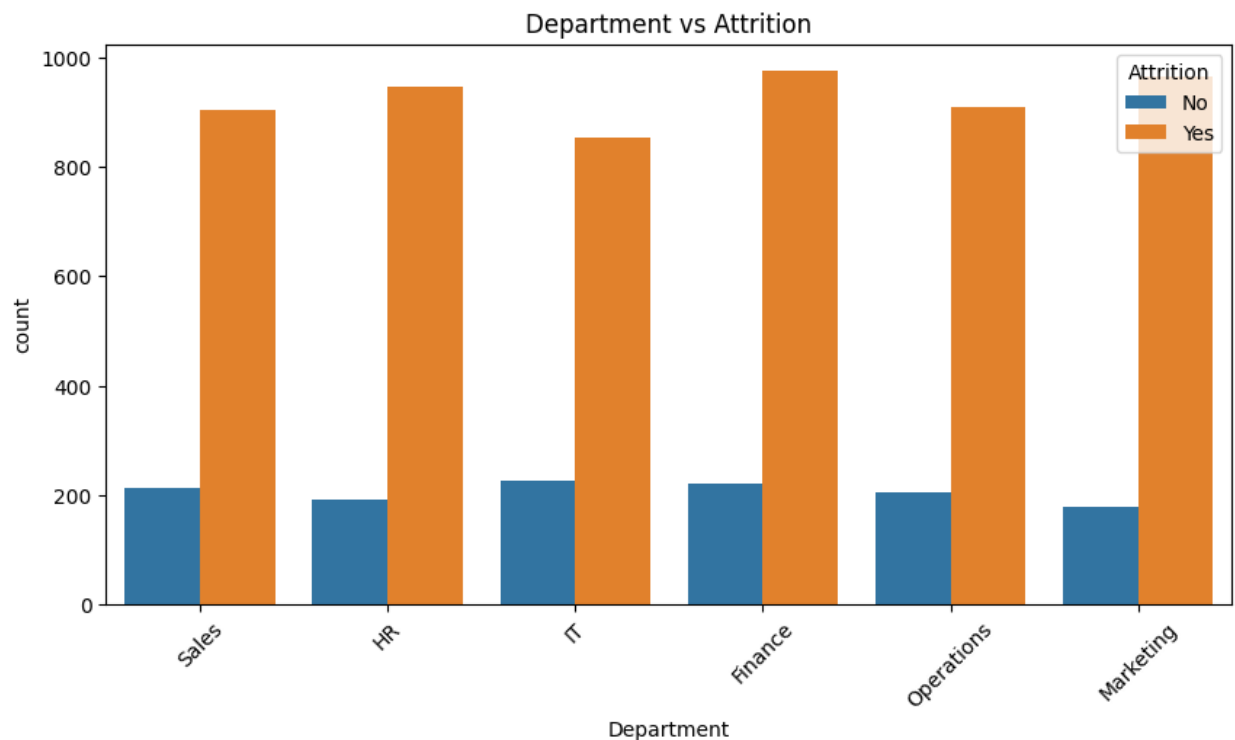
Performance Rating Group

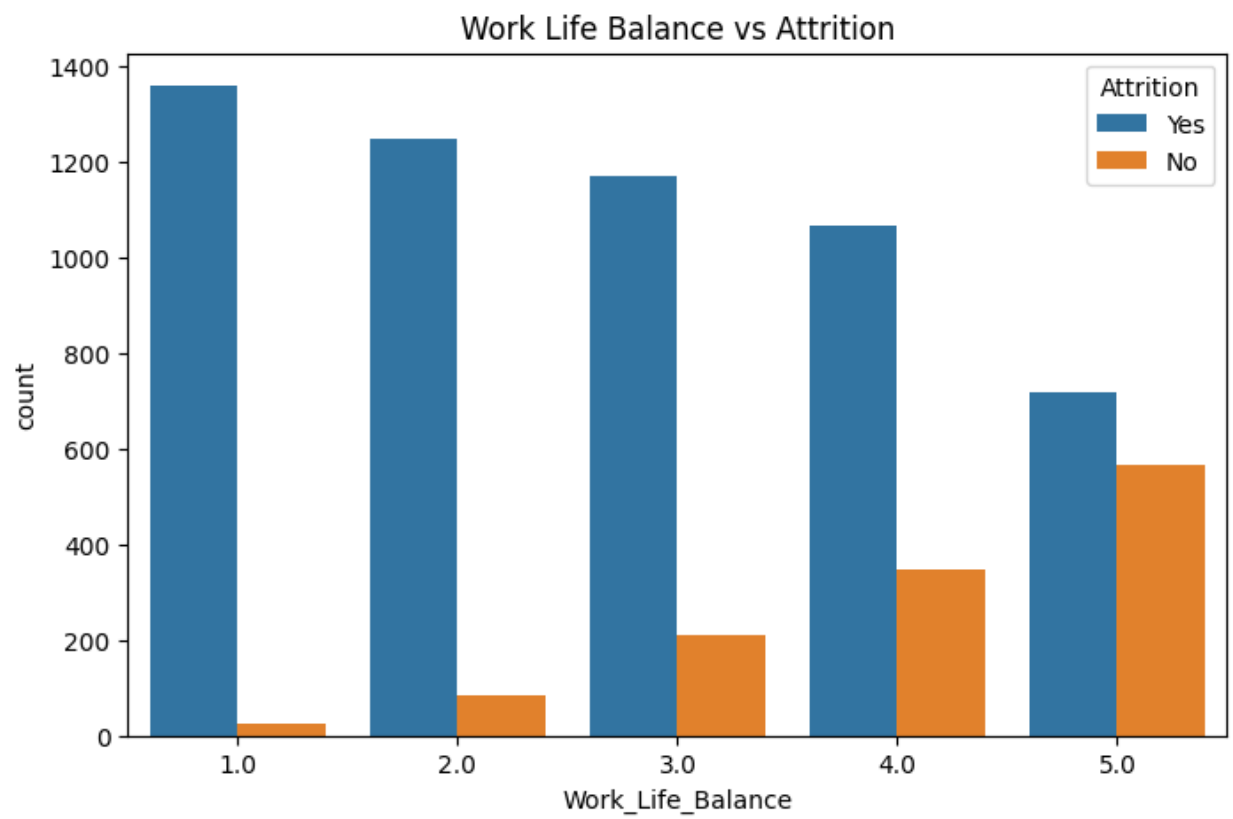
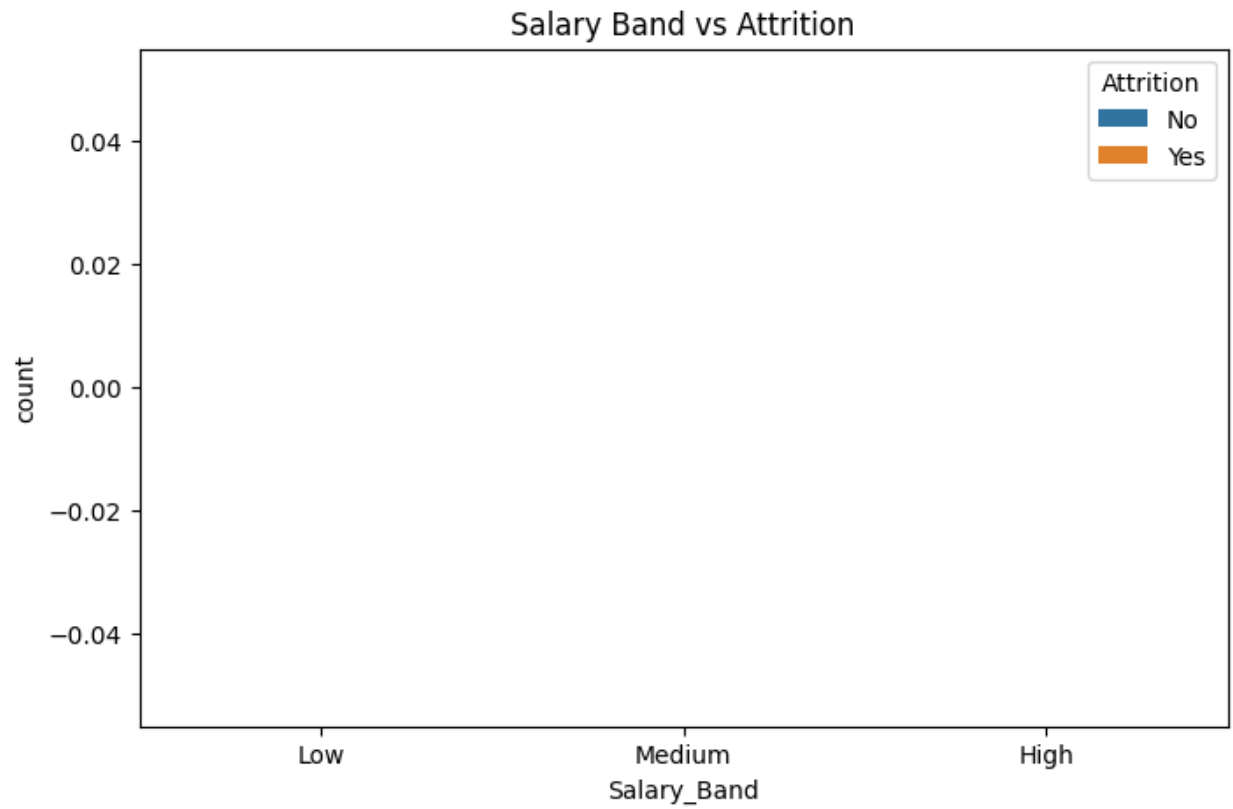
Performance ratings were categorized for analysis.

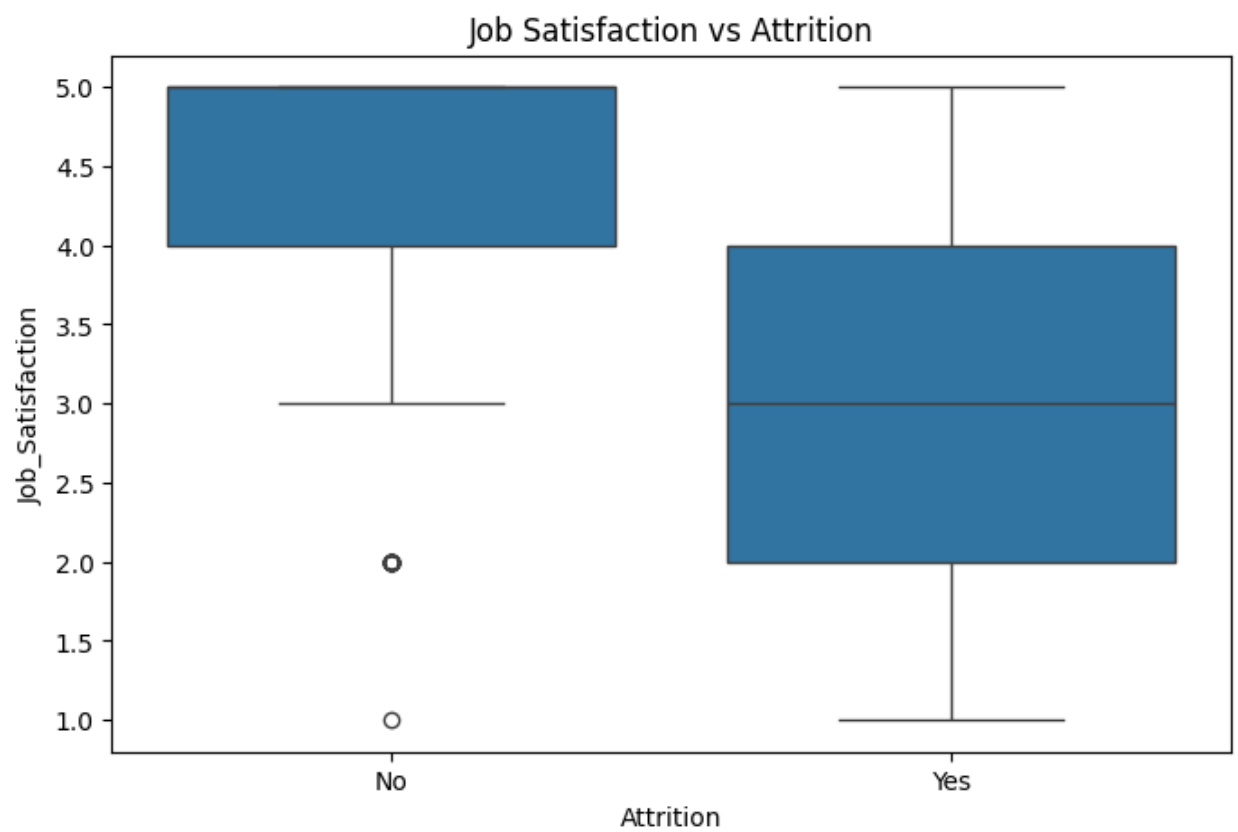
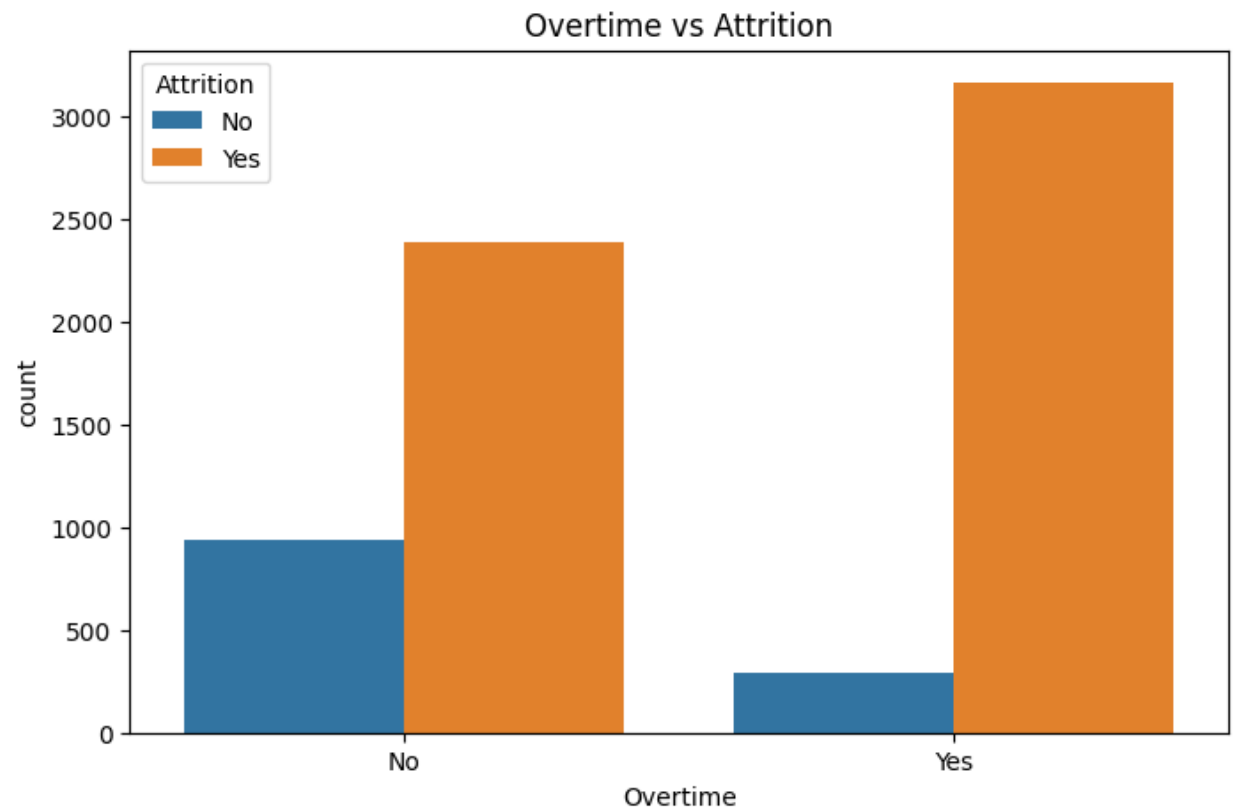
Employee Segmentation

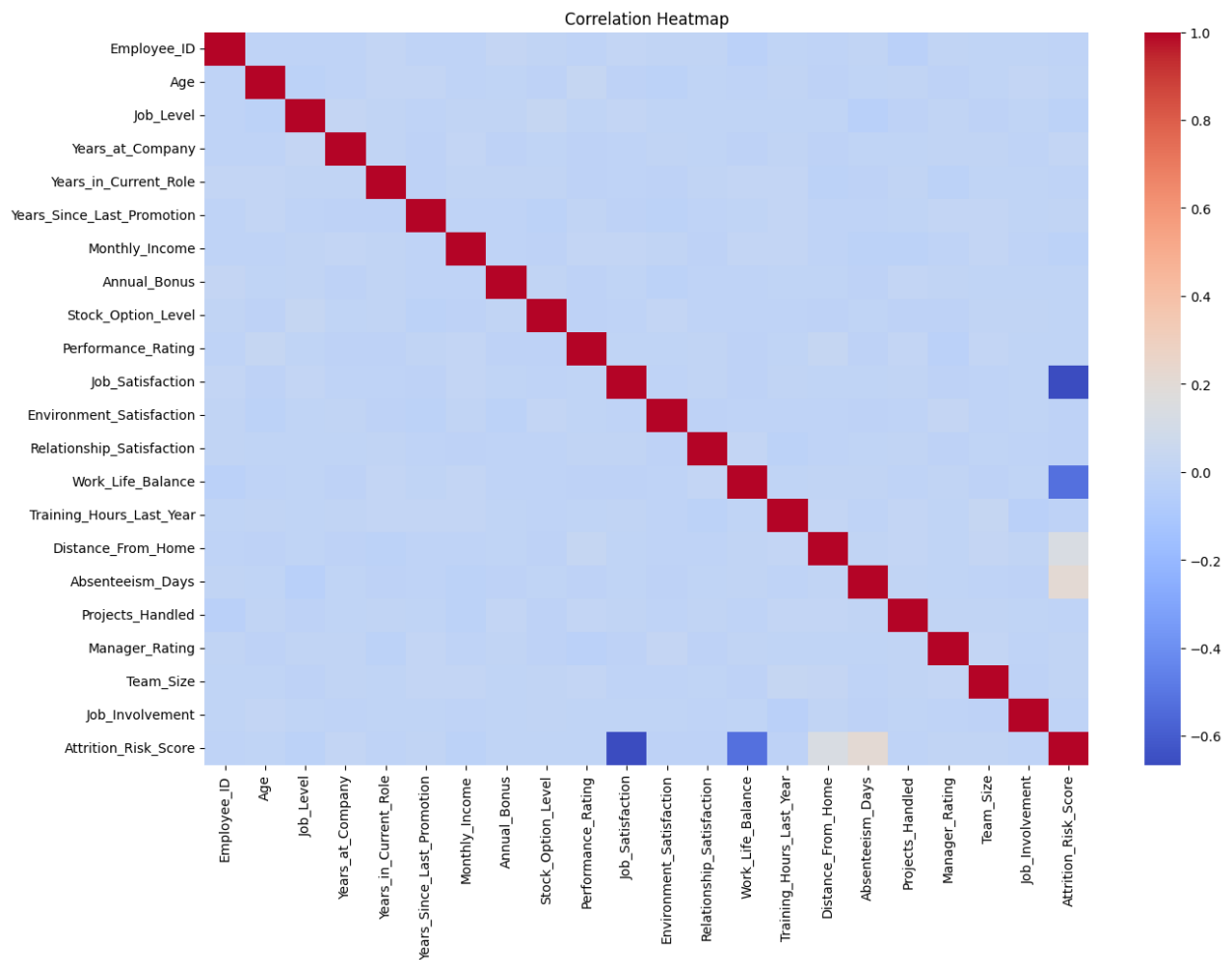
Employees were segmented into:

- Overtime Employees
- High Performers
- Long-Term Employees
- Dissatisfied Employees









7. KPI Design & Metrics

The following Key Performance Indicators (KPIs) were designed.

Total Employees

Measures the total workforce size.

Attrition Count

Measures the number of employees leaving the organization.

Attrition Rate

Attrition Rate = (Attrition Count / Total Employees) × 100

Retention Rate

Retention Rate = $100 - \text{Attrition Rate}$

Average Job Satisfaction

Measures employee satisfaction levels.

Average Environment Satisfaction

Measures satisfaction with workplace conditions.

Average Work-Life Balance

Measures employees' ability to balance professional and personal life.

Average Performance Rating

Measures overall employee performance.

Overtime Percentage

Measures workforce overtime prevalence.

8. Dashboard Development

Four interactive dashboards were developed.

Executive Overview Dashboard**Purpose**

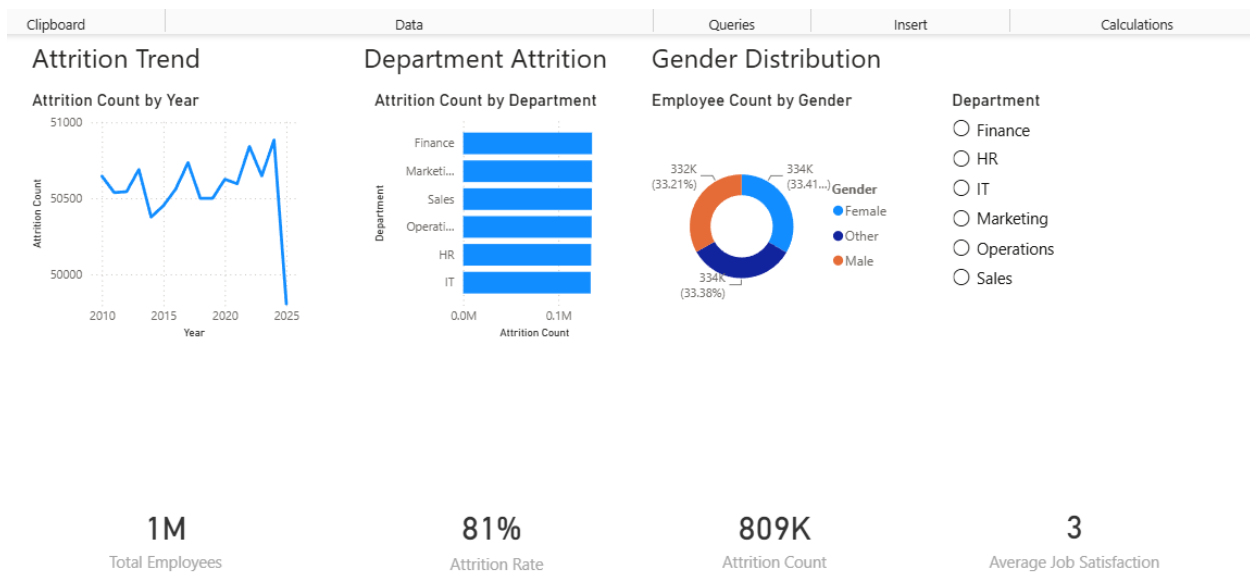
Provide a high-level overview of workforce health.

Visuals Included

- Total Employees KPI
- Attrition Count KPI
- Attrition Rate KPI
- Retention Rate KPI
- Attrition Trend Analysis

Business Value

Supports executive-level workforce monitoring.



Workforce Dashboard

Purpose

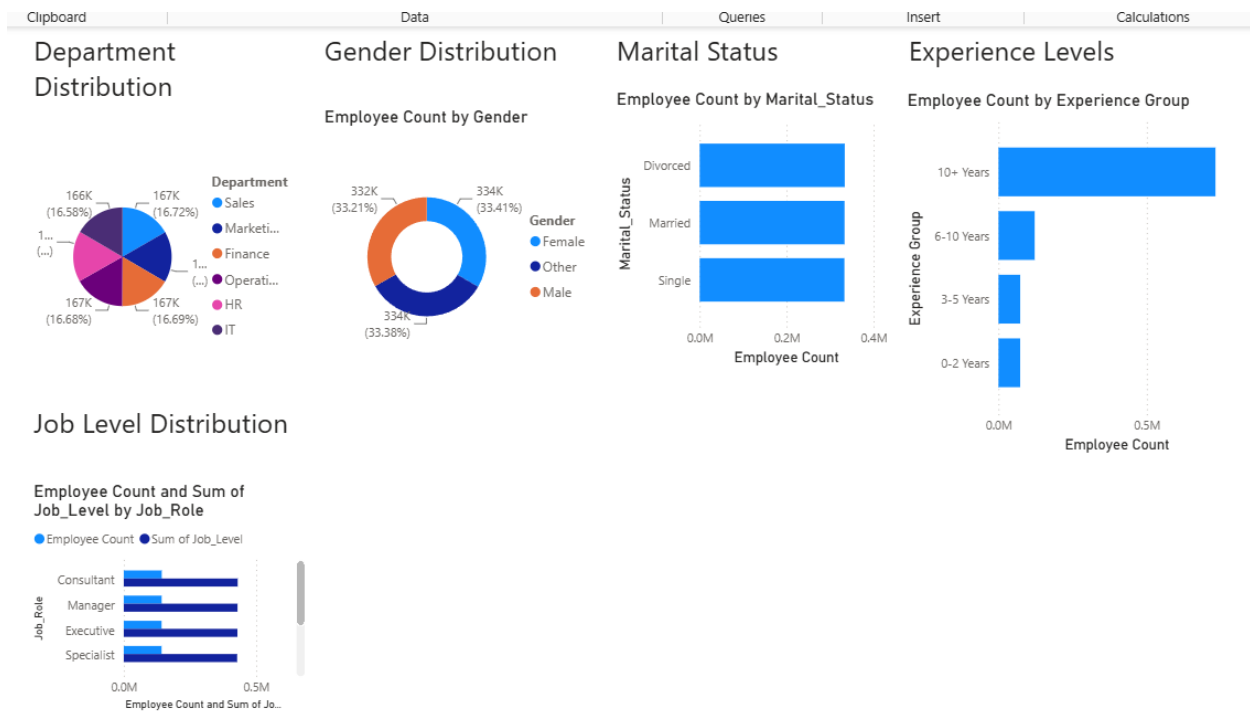
Analyze workforce composition.

Visuals Included

- Department Distribution
- Gender Distribution
- Marital Status Distribution
- Experience Level Distribution
- Job Role Distribution

Business Value

Provides insights into workforce demographics.



Attrition Dashboard

Purpose

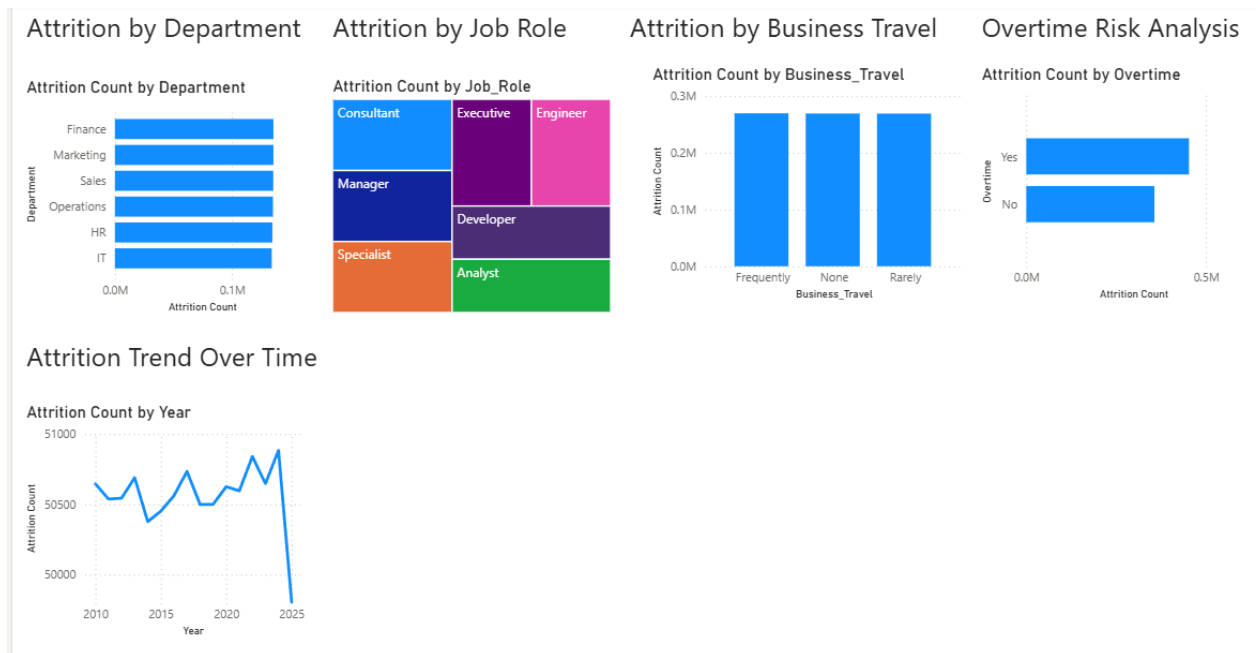
Analyze employee turnover patterns.

Visuals Included

- Attrition by Department
- Attrition by Job Role
- Attrition by Business Travel
- Overtime Risk Analysis
- Attrition Trend Over Time

Business Value

Helps identify workforce segments with higher attrition.



Employee Health Dashboard

Purpose

Monitor employee engagement and well-being.

Visuals Included

- Job Satisfaction Analysis
- Environment Satisfaction Analysis
- Overtime Analysis
- Work-Life Balance Analysis
- Performance Rating Analysis

Business Value

Supports proactive workforce management.



9. Analytics & Findings

Workforce Insights

The workforce is distributed across Finance, HR, IT, Marketing, Operations, and Sales departments.

Employee demographics show balanced representation across different workforce groups.

Satisfaction Analysis

Employee satisfaction indicators are generally moderate, suggesting opportunities for engagement improvement.

Work-life balance scores indicate the need for continued employee wellness initiatives.

Overtime Analysis

A significant portion of employees perform overtime work.

Excessive overtime may contribute to employee fatigue and increased attrition risk.

Attrition Insights

Attrition varies across departments and employee groups.

Business travel frequency and overtime appear to influence employee turnover patterns.

Performance Analysis

Employee performance ratings remain relatively stable across the workforce.

Performance metrics can be used alongside satisfaction indicators to identify high-potential employees.

10. Business Recommendations

Recommendation 1

Reduce excessive overtime through workload balancing.

Recommendation 2

Implement employee wellness and engagement programs.

Recommendation 3

Develop department-specific retention initiatives.

Recommendation 4

Improve career development and growth opportunities.

Recommendation 5

Conduct regular employee feedback surveys.

Recommendation 6

Strengthen work-life balance policies and flexible work arrangements.

11. Conclusion

The HR Workforce Attrition Analysis project successfully developed an interactive dashboard solution for monitoring workforce health, employee engagement, and attrition patterns.

The dashboards provide valuable insights into employee behavior, workforce composition, and organizational risk factors. These insights enable HR leaders to make informed decisions that improve retention and workforce stability.

The project demonstrates the application of HR Analytics, Business Intelligence, and Data Visualization techniques to solve real-world workforce challenges.

12. Future Scope

Future enhancements may include:

- Predictive Attrition Modeling
- Employee Churn Prediction
- Machine Learning-Based Risk Scoring
- Employee Sentiment Analysis
- Real-Time Dashboard Integration
- AI-Powered Retention Recommendations

These improvements can further strengthen workforce planning and strategic HR decision-making.

Questions:

1. Why are employees leaving?

Employees leave due to factors such as low job satisfaction, limited career growth, poor work-life balance, and inadequate compensation. Analyzing employee feedback and exit data helps identify the main causes.

2. Which departments experience higher attrition?

Some departments may have higher turnover rates because of workload, job stress, or management issues. Department-wise attrition analysis helps identify areas that need attention.

3. Are salary and work-life balance affecting retention?

Employees with lower salaries or poor work-life balance are often more likely to leave. Comparing attrition rates across salary levels and work-life balance ratings reveals their impact on retention.

4. Which employee groups are at risk?

Groups such as new hires, employees working overtime, or those with low job satisfaction may have a higher risk of leaving. Identifying these groups enables proactive retention efforts.

5. How can organizations improve retention?

Organizations can improve retention by offering competitive salaries, career development opportunities, flexible work arrangements, and employee recognition programs. These initiatives increase satisfaction and reduce turnover.